A positive answer to the real-line transitive Lipschitz question

Run fa_banach_001, agent_lane_sp

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Source Question

The source paper is Abbar, Coine, and Petitjean, *On the dynamics of Lipschitz operators*, arXiv:2011.10800. Remark 4.4 asks whether the compact-interval theorem proved there remains true on the real line:

> We do not know whether $\widehat{f} : \mathcal{F}(\mathbb{R}) \to \mathcal{F}(\mathbb{R})$ is hypercyclic whenever $f : \mathbb{R} \to \mathbb{R}$ is a transitive Lipschitz map having a fixed point.

The remark explains that the source proof for compact intervals used a Barge-Martin/Sy decomposition and says that no comparable result was known for unbounded intervals.

> *Remark* 4.4. We do not know whether $\widehat{f} : \mathcal{F}(\mathbb{R}) \to \mathcal{F}(\mathbb{R})$ is hypercyclic whenever $f : \mathbb{R} \to \mathbb{R}$ is a transitive Lipschitz map (having a fixed point). In fact, the decomposition given by [26, Theorem 2.19] was crucial to our proof (see also [5, Theorem 2.2] for a similar statement in the more general setting of locally connected compact metric spaces). We do not know if a comparable result holds in the case of unbounded intervals. Nevertheless, any weakly mixing Lipschitz maps $f : \mathbb{R} \to \mathbb{R}$ produces a weakly mixing Lipschitz operator acting on $L^1(\mathbb{R})$ thanks to [23, Theorem 2.3] (see [24] at page 2 for an example).

Answer

Theorem 1. *Let $f : \mathbb{R} \to \mathbb{R}$ be Lipschitz and topologically transitive. Suppose that f has a fixed point z, and point $\mathbb{R}$ at z. Then the induced operator*

$$\widehat{f} : \mathcal{F}_z(\mathbb{R}) \longrightarrow \mathcal{F}_z(\mathbb{R})$$

is weakly mixing. In particular, $\widehat{f}$ is hypercyclic.

Moreover, $\widehat{f}$ has dense periodic points, hence it is Devaney chaotic.

External Inputs

We use three standard facts.

(a) The source paper recalls the theorem of Murillo-Arcila and Peris: if a Lipschitz self-map $h : M \to M$ is weakly mixing, mixing, or weakly mixing and chaotic, then its Lipschitz-free linearization $\widehat{h}$ has the corresponding linear dynamical property on $\mathcal{F}(M)$.

(b) The source paper's Proposition 2.6 says that if $\mathrm{Per}(h)$ is dense in M, then $\mathrm{Per}(\widehat{h})$ is dense in $\mathcal{F}(M)$.

(c) Ansari's power theorem says that, for a bounded linear operator S, hypercyclicity of a positive power S^m implies hypercyclicity of S.

We also use two theorems of Du, *Continuous transitive maps on the interval revisited*, arXiv:1701.02589. Du's Theorem 4 applies to every real interval, including $\mathbb{R}$. For a continuous transitive interval map, either a same-side fixed-point condition holds, or there is a unique fixed point and the square of the map is transitive on the two sides; in the second case the restricted square maps have at least two fixed points. In all cases, Du's Theorem 4 gives dense periodic points.

Theorem 4. *Let I denote one of the following intervals: (a,b), $(a,b]$, $[a,b)$, $[a,b]$, $(-\infty,b)$, $(-\infty,b]$, (a,∞), $[a,\infty)$ and $(-\infty,\infty)$, where $a<b$ are real numbers, and let f be a continuous transitive map on I, or, equivalently, assume that there is a point u in I whose orbit $O_f(u)$ is dense in I. Then exactly one of the following holds:*

(1) *If, for some fixed point z of f and some point $c \neq z$, we have $\big(f(c)-z\big)/(c-z) > 0$, then, for any compact interval K in I and any compact interval L in the interior of I, there exists a positive integer N such that $f^n(K) \supset L$ for all $n \geq N$.*

(2) *There exists a unique fixed point z of f in the interior of I such that $\big(f(x)-z\big)/(x-z) \leq 0$ for all $x \neq z$ in I and, on each of the intervals $I \cap (-\infty, z]$ and $I \cap [z,\infty)$, f^2 is transitive and has*

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at least two fixed points. Consequently, in this case $\big($i.e., f is transitive but not bitransitive on $I\big)$, I must be a compact interval or an open interval in the real line.

Consequently, if f is transitive on I, then f has dense periodic points in I.

Du's Theorem 6 says that, on intervals including closed half-lines, the same-side fixed-point condition is equivalent to mixing and weak mixing.

Theorem 6. *Let I denote one of the intervals: (a,b), $(a,b]$, $[a,b)$, $[a,b]$, $(-\infty,b)$, $(-\infty,b]$, (a,∞), $[a,\infty)$ and $(-\infty,\infty)$, where $a<b$, are real numbers, except in statements (3), (4) & (5) where I denotes one of $[a,b]$, $[a,\infty)$, $(-\infty,b]$, $(-\infty,\infty)$, and let $f: I \to I$ be a continuous map. Then the following statements are equivalent:*

(1) *f is transitive on I and there exist a fixed point z of f and a point $c \neq z$ such that either $z < \min\{c, f(c)\}$ or $\max\{c, f(c)\} < z$, i.e., both points c and $f(c)$ lie on the same side of z.*

(2) *For any compact subinterval K of I and any compact subinterval L of $int(I)$, there exists a positive integer M such that $f^n(K) \supset L$ for all $n \geq M$. Consequently, for any positive integer m, any compact intervals K_i, $1 \leq i \leq m$, and any intervals V_i, $1 \leq i \leq m$, in I, by taking, for each $1 \leq i \leq m$, a compact interval L_i in the interior of V_i, we obtain that there exists a positive integer N_m such that $f^n(K_i) \supset L_i$ for all $1 \leq i \leq m$ and all $n \geq N_m$, and so, there exist compact intervals $K^*_{n,i}$ in K_i, $1 \leq i \leq m$, which satisfy that*

$$f^n(K^*_{n,i}) \subset L_i \subset V_i \text{ for all } 1 \leq i \leq m \text{ and all } n \geq N_m.$$

(We phrase it this way for later use in the proof of Theorem 8 below. Note that since we only require the set $f^n(K^*_{n,i})$ to be contained in V_i, we can choose the compact interval $K^*_{n,i}$ as small as we wish).

(3) *For any open interval U in I, there exists a positive integer $m(U)$ such that f has a period-n point in U for each integer $n \geq m(U)$ and f has no proper invariant closed intervals in I.*

(4) *For some infinite set $\mathcal{N}$ of positive integers which contains an odd integer ≥ 3, the periodic points of f with periods $n \in \mathcal{N}$ are dense in I and f has no proper invariant closed intervals in I.*

(5) *The set of periodic points of f is dense and f^2 has no proper invariant closed interval in I.*

(6) *f is (topologically) mixing, i.e., for any two non-empty open subsets U and V of I, there exists a positive integer $m(U,V)$ such that $f^n(U) \cap V \neq \emptyset$ for all $n \geq m(U,V)$.*

(7) *f is (topologically) weakly mixing, i.e., the map $f \times f$ defined by $(f \times f)((x,y)) = (f(x), f(y))$ is transitive on $I \times I$.*

(8) *For any infinite subset $\mathcal{M}$ of the positive integers, there is a residual set S (i.e., a set containing the intersection of countably infinitely many open dense sets) in I such that S is dense in I and if $x \in S$ then the set $\{f^n(x) : n \in \mathcal{M}\}$ is dense in I.*

(9) *f^{2k} is transitive on I for some positive integer k.*

(10) *f is totally transitive, i.e., f^k is transitive on I for each integer $k \geq 1$.*

Idea of the Proof

The source proof for compact intervals splits the free space according to the two invariant halves of the square map. The only missing ingredient for the real line is the corresponding topological decomposition. Du's Theorem 4 supplies it.

If f satisfies Du's same-side fixed-point alternative, then f is mixing by Du's Theorem 6, so $\widehat{f}$ is mixing by the Murillo-Arcila–Peris implication recalled in the source paper.

Otherwise f has a unique fixed point z, swaps the two half-lines, and f^2 is transitive on each half-line with at least two fixed points. Du's Theorem 6 then makes each half-line restriction of f^2 mixing. Under the standard identification $\mathcal{F}_z(\mathbb{R}) \cong L^1(\mathbb{R})$, the two half-lines are the two L^1-summands, so $\widehat{f^2}$ is a direct sum of two mixing operators. This makes $\widehat{f^2}$ mixing, and hence $\widehat{f}$ is weakly mixing by Ansari's power theorem.

Finally, Du gives dense periodic points for f, and the source paper's periodic-point proposition transfers this density to $\widehat{f}$.

Proof

Let $T = \widehat{f}$. Because $f(z) = z$, T is a bounded operator on the pointed Lipschitz-free space $\mathcal{F}_z(\mathbb{R})$.

Apply Du's Theorem 4 to $f : \mathbb{R} \to \mathbb{R}$. There are two cases.

Case 1: the same-side alternative. There are a fixed point p of f and a point $c \neq p$ such that

$$\frac{f(c) - p}{c - p} > 0.$$

Equivalently, c and $f(c)$ lie on the same side of the fixed point p. By Du's Theorem 6, f is mixing, hence weakly mixing. Therefore $T = \widehat{f}$ is mixing, in particular weakly mixing, by the Murillo-Arcila–Peris transfer theorem recalled in the source paper.

Case 2: the side-swapping alternative. Du's Theorem 4 gives a unique fixed point z and

$$f((-\infty, z]) \subseteq [z, \infty), \qquad f([z, \infty)) \subseteq (-\infty, z],$$

so $g = f^2$ preserves the two closed half-lines

$$I_- := (-\infty, z], \qquad I_+ := [z, \infty).$$

It also says that the restrictions

$$g_- := g|_{I_-} : I_- \to I_-, \qquad g_+ := g|_{I_+} : I_+ \to I_+$$

are transitive and have at least two fixed points.

Each $I_\pm$ is one of the half-line intervals covered by Du's Theorem 6. Since $g_\pm$ has two fixed points, choose one as p and the other as c. Then $g_\pm(c) = c$, so c and $g_\pm(c)$ are on the same side of p. Thus condition (1) of Du's Theorem 6 holds for $g_\pm$, and g_- and g_+ are mixing.

By the Murillo-Arcila–Peris transfer theorem, the induced operators

$$\widehat{g_-} : \mathcal{F}_z(I_-) \to \mathcal{F}_z(I_-), \qquad \widehat{g_+} : \mathcal{F}_z(I_+) \to \mathcal{F}_z(I_+)$$

are mixing.

Now use the standard real-tree/free-space identification. With base point z,

$$\Phi : \mathcal{F}_z(\mathbb{R}) \longrightarrow L^1(\mathbb{R}), \qquad \Phi(\delta_x) = \begin{cases} \mathbf{1}_{[z,x]}, & x \geq z, \\ -\mathbf{1}_{[x,z]}, & x < z, \end{cases}$$

is an isometric isomorphism. Under Φ, the subspaces $\mathcal{F}_z(I_-)$ and $\mathcal{F}_z(I_+)$ are exactly the two summands

$$L^1((-\infty, z]) \oplus_1 L^1([z, \infty)) = L^1(\mathbb{R}).$$

Since $g = f^2$ preserves the two half-lines, $T^2 = \widehat{f^2}$ is, under this decomposition, the direct sum of $\widehat{g_-}$ and $\widehat{g_+}$. A direct sum of two mixing operators is mixing: given open balls in the ℓ_1-sum, refine them to product balls in the two summands and use the same large iterate for both mixing restrictions.

Thus T^2 is mixing. Consequently $(T \oplus T)^2 = T^2 \oplus T^2$ is hypercyclic. By Ansari's power theorem, $T \oplus T$ is hypercyclic. This is equivalent to weak mixing of T. Hence $T = \widehat{f}$ is weakly mixing in Case 2 as well.

In both cases T is weakly mixing, and therefore hypercyclic.

It remains only to record the stronger Devaney-chaos conclusion. Du's Theorem 4 also states that every continuous transitive interval map has dense periodic points. Therefore $\mathrm{Per}(f)$ is dense in $\mathbb{R}$. By the source paper's Proposition 2.6, $\mathrm{Per}(\widehat{f})$ is dense in $\mathcal{F}_z(\mathbb{R})$. Since $T = \widehat{f}$ is already hypercyclic, T is Devaney chaotic. □

Verification Notes and Limitations

This proves the question in Remark 4.4 as stated, after the standard choice of a fixed point as the base point of $\mathbb{R}$. If the source paper's notation $\mathcal{F}(\mathbb{R})$ is read with a different fixed base point, translate the real line so that the chosen fixed point becomes the base point; Lipschitz-free spaces over pointed translates are canonically isometric.

The proof does not answer the separate question in the source paper asking whether a Lipschitz operator can be hypercyclic while failing the Hypercyclicity Criterion. In fact, the conclusion obtained here is stronger than hypercyclicity: $\widehat{f}$ is weakly mixing, so in the linear setting it satisfies the Hypercyclicity Criterion.

Novelty Check

The run's cheap indexes had no prior packet for arXiv:2011.10800 or this real-line transitive-map question. The adjacent indexed packets for arXiv:2310.08965 and arXiv:2512.05640 concern different Lipschitz-free operator problems.

On 2026-06-30, web/arXiv searches for phrases including "transitive Lipschitz map", "F(R)", "widehat f", "hypercyclic", and "unbounded intervals" did not find a later paper explicitly answering Remark 4.4. The proof uses Du's earlier interval-dynamics theorem as the missing decomposition, so the result should be reviewed as a likely-valid operator consequence of known interval dynamics rather than as a new theorem about transitive interval maps.

Human Review Recommendation

Review the use of Du's Theorem 6 on the two closed half-lines, and verify the standard $\mathcal{F}_z(\mathbb{R}) \cong L^1(\mathbb{R})$ splitting is acceptable in the source paper's convention. If those two checks pass, this is a full positive answer to Remark 4.4.

References

[1] A. Abbar, C. Coine, and C. Petitjean, *On the dynamics of Lipschitz operators*, arXiv:2011.10800.

[2] B.-S. Du, *Continuous transitive maps on the interval revisited*, arXiv:1701.02589.

[3] M. Murillo-Arcila and A. Peris, *Chaotic behaviour on invariant sets of linear operators*, Integral Equations and Operator Theory 81 (2015), 483–497.

[4] S. I. Ansari, *Hypercyclic and cyclic vectors*, Journal of Functional Analysis 128 (1995), 374–383.